Sanity Check: Comparison of Baseline-corrected FGEs (E1; R output, JASP, Excel)

R

Baseline-Corrected PSE Comparisons for Within-Version Factors

Note. inflated p-values due to Bonferroni correction

Table D1

Baseline-Corrected PSE Comparisons Within Motion Type and Spatial Location

Comparison	M_1 (*SD*)	M_2 (*SD*)	*t*	*p*	*d*	BF_{01}
Orth: FP vs FF	0.9 (0.6)	0.9 (0.3)	0.22	11.63	0.04	4.56
Rad: FP vs FF	0.5 (0.6)	0.7 (0.4)	-1.43	2.32	-0.29	1.89
Orth FP: L vs R	0.9 (0.7)	0.8 (0.8)	0.49	8.79	0.10	4.17
Orth FP: UP vs LO	0.7 (0.7)	1.0 (0.7)	-2.48	0.29	-0.51	0.38
Orth FP: IN vs OUT	0.4 (0.7)	1.5 (1.1)	-5.05	<.001***	-1.03	0.00
Rad FP: L vs R	0.6 (0.8)	0.4 (0.8)	0.97	4.77	0.20	3.04
Rad FP: UP vs LO	0.5 (0.8)	0.6 (0.8)	-0.59	7.85	-0.12	3.98
Rad FP: IN vs OUT	-0.2 (0.8)	1.2 (1.1)	-7.18	<.001***	-1.47	0.00
Orth FF: L vs R	0.9 (0.4)	0.9 (0.4)	0.36	10.07	0.07	4.39
Orth FF: UP vs LO	1.0 (0.4)	0.8 (0.4)	1.82	1.14	0.37	1.12
Orth FF: IN vs OUT	1.3 (0.5)	0.4 (0.5)	5.50	<.001***	1.12	0.00
Rad FF: L vs R	0.7 (0.6)	0.7 (0.5)	0.12	12.71	0.02	4.63
Rad FF: UP vs LO	0.7 (0.5)	0.7 (0.6)	-0.01	13.92	-0.00	4.66
Rad FF: IN vs OUT	1.5 (0.9)	0.1 (0.5)	6.53	<.001***	1.33	0.00

Note. PSEs represent baseline-corrected FGE. Petal FGE values have been sign-reversed prior to comparison. Positive values indicate a larger mislocalisation consistent with the direction of post-flash motion. Cohen's *d* represents effect size for paired samples. BF_{01} = Bayes factor favoring null hypothesis. * p < .05, ** p < .01, *** p < .001.

JASP

Paired Samples T-Test

Measure 1		Measure 2	t	df	p
Petal_O_FGE	-	Fugal_O_FGE	0.2162459946	23	.8307024415
Petal_R_FGE	-	Fugal_R_FGE	−1.4321289855	23	.1655580822
Petal_O_Left_FGE	-	Petal_O_Right_FGE	0.4914369793	23	.6277745111
Fugal_O_Left_FGE	-	Fugal_O_Right_FGE	0.3642866007	23	.7189717347
Petal_R_Left_FGE	-	Petal_R_Right_FGE	0.9735665987	23	.3404004703
Fugal_R_Left_FGE	-	Fugal_R_Right_FGE	0.1171919033	23	.9077253473
Petal_O_Lower_FGE	-	Petal_O_Upper_FGE	2.4802758142	23	.0208817802
Fugal_O_Lower_FGE	-	Fugal_O_Upper_FGE	−1.8241420982	23	.0811508752
Petal_R_Lower_FGE	-	Petal_R_Upper_FGE	0.5908782566	23	.5603643020
Fugal_R_Lower_FGE	-	Fugal_R_Upper_FGE	0.0071009150	23	.9943955699
Petal_O_Inner_FGE	-	Petal_O_Outer_FGE	−5.0525588318	23	.0000409484
Fugal_O_Inner_FGE	-	Fugal_O_Outer_FGE	5.5036788751	23	.0000134912
Petal_R_Inner_FGE	-	Petal_R_Outer_FGE	−7.1811452981	23	.0000002597
Fugal_R_Inner_FGE	-	Fugal_R_Outer_FGE	6.5286745752	23	.0000011609

Note. Student's t-test.

Bayesian Paired Samples T-Test

Measure 1		Measure 2	BF_{01}	error %
Petal_O_FGE	-	Fugal_O_FGE	4.5600124313	0.0239817058
Petal_R_FGE	-	Fugal_R_FGE	1.8923747726	0.0257733483
Petal_O_Left_FGE	-	Petal_O_Right_FGE	4.1731047793	0.0244568942
Fugal_O_Left_FGE	-	Fugal_O_Right_FGE	4.3846407182	0.0242006587
Petal_R_Left_FGE	-	Petal_R_Right_FGE	3.0436373037	0.0255853850
Fugal_R_Left_FGE	-	Fugal_R_Right_FGE	4.6294356903	0.0238935954
Petal_O_Lower_FGE	-	Petal_O_Upper_FGE	0.3805993198	0.0000009655
Fugal_O_Lower_FGE	-	Fugal_O_Upper_FGE	1.1193485880	0.0242255792
Petal_R_Lower_FGE	-	Petal_R_Upper_FGE	3.9747832496	0.0246874376
Fugal_R_Lower_FGE	-	Fugal_R_Upper_FGE	4.6585350474	0.0238562953
Petal_O_Outer_FGE	-	Petal_O_Inner_FGE	0.0016542936	0.0000000146
Fugal_O_Inner_FGE	-	Fugal_O_Outer_FGE	0.0006048304	0.0000000128
Petal_R_Inner_FGE	-	Petal_R_Outer_FGE	0.0000160945	0.0000000025
Fugal_R_Inner_FGE	-	Fugal_R_Outer_FGE	0.0000640999	0.0000000005

EXCEL

FGE (PETAL vs FUGAL) [MINUS CONTROL]

[note petal values have been converted to positive prior to t.test]

ORTHOGONAL			RADIAL		
PETAL VS. FUGAL			PETAL VS FUGAL		
means	-0.884	0.859	means	-0.522	0.7203
p =	0.8307		p =	0.1656	
PETAL LEFT VS PETAL RIGHT			PETAL LEFT VS PETAL RIGHT		
means	-0.905	-0.834	means	-0.612	-0.44
p =	0.3139		p =	0.3404	
FUGAL LEFT VS FUGAL RIGHT			FUGAL LEFT VS FUGAL RIGHT		
means	0.8896	0.8512	means	0.7378	0.7204
p =	0.719		p =	0.9077	
PETAL UPPER vs PETAL LOWER			PETAL UPPER vs PETAL LOWER		
means	-0.745	-1.023	means	-0.475	-0.563
p =	0.0209		p =	0.5604	
FUGAL UPPER vs FUGAL LOWER			FUGAL UPPER vs FUGAL LOWER		
means	0.9616	0.7633	means	0.7379	0.7387
p =	0.0812		p =	0.9944	
PETAL INNER vs PETAL OUTER			PETAL INNER vs PETAL OUTER		
means	-0.392	-1.5	means	0.2224	-1.183
p =	<.001		p =	<.001	
FUGAL INNER vs FUGAL OUTER			FUGAL INNER vs FUGAL OUTER		
means	1.2571	0.3676	means	1.4501	0.0647
p =	<.001		p =	<.001	

FGE COMPARISONS (AVERAGED OVER PETAL & FUGAL) [MINUS CONTROL]

[note petal values have been converted to positive prior to t.test] - these have not been calculated as PSEs because of the sign reversal prior to averaging.

ORTHOGONAL			RADIAL		
LEFT vs RIGHT			LEFT VS RIGHT		
means	0.8975	0.8425	means	0.6751	0.5801
p =	0.5395		p =	0.4132	
UPPER vs LOWER			UPPER vs LOWER		
means	0.8535	0.8933	means	0.6067	0.6507
p =	0.6407		p =	0.6392	
INNER vs OUTER			INNER vs OUTER		
means	0.6139	0.6236	means	0.8246	0.9339
p =	0.9688		p =	0.5846	

Sanity Check: Comparison of Raw PSEs (E1; R output, JASP)

R

Raw PSE Comparisons for Within-Version Factors

Table C1

Raw PSE Comparisons Within Motion Type and Spatial Location

Comparison	M_1 (*SD*)	M_2 (*SD*)	*t*	*p*	*d*	BF_{01}
Orth: FP vs FF	2.2 (0.8)	-0.5 (0.6)	11.10	<.001***	2.27	0.00
Rad: FP vs FF	1.7 (1.0)	-0.4 (0.7)	6.75	<.001***	1.38	0.00
Orth FP: L vs R	2.2 (0.9)	2.2 (1.0)	0.37	9.99	0.08	4.37
Orth FP: UP vs LO	2.3 (1.0)	2.1 (0.9)	2.17	0.57	0.44	0.65
Orth FP: IN vs OUT	1.8 (1.0)	2.7 (1.2)	-4.36	0.003**	-0.89	0.01
Rad FP: L vs R	1.7 (0.9)	1.6 (1.1)	0.78	6.22	0.16	3.54
Rad FP: UP vs LO	1.9 (1.2)	1.5 (1.0)	2.59	0.23	0.53	0.31
Rad FP: IN vs OUT	1.3 (1.0)	2.1 (1.5)	-3.28	0.05*	-0.67	0.08
Orth FF: L vs R	-0.4 (0.6)	-0.5 (0.6)	0.52	8.55	0.11	4.13
Orth FF: UP vs LO	-0.6 (0.6)	-0.3 (0.6)	-3.23	0.05	-0.66	0.09
Orth FF: IN vs OUT	-0.2 (0.6)	-0.9 (0.7)	6.34	<.001***	1.29	0.00
Rad FF: L vs R	-0.3 (0.7)	-0.4 (0.8)	0.57	8.01	0.12	4.01
Rad FF: UP vs LO	-0.6 (0.6)	-0.2 (0.9)	-3.41	0.03*	-0.70	0.06
Rad FF: IN vs OUT	-0.0 (0.8)	-0.8 (1.0)	3.76	0.01*	0.77	0.03

JASP

Paired Samples T-Test ▼

Measure 1		Measure 2	t	df	p
Petal_Orthogonal	-	Fugal_Orthogonal	11.0989138985	23	.0000000001
Petal_Radial	-	Fugal_Radial	6.7499751746	23	.0000006944
Petal_O_Left	-	Petal_O_Right	0.3714062502	23	.7137344197
Petal_O_Upper	-	Petal_O_Lower	2.1724177872	23	.0403799987
Petal_O_Inner	-	Petal_O_Outer	−4.3579345504	23	.0002307126
Petal_R_Left	-	Petal_R_Right	0.7780468971	23	.4444719985
Petal_R_Lower	-	Petal_R_Upper	−2.5901419393	23	.0163668666
Petal_R_Inner	-	Petal_R_Outer	−3.2786267585	23	.0032953518
Fugal_O_Left	-	Fugal_O_Right	0.5163597897	23	.6105323760
Fugal_O_Upper	-	Fugal_O_Lower	−3.2330945022	23	.0036758025
Fugal_O_Inner	-	Fugal_O_Outer	6.3351347649	23	.0000018285
Fugal_R_Left	-	Fugal_R_Right	0.5727145822	23	.5723947688
Fugal_R_Upper	-	Fugal_R_Lower	−3.4138591482	23	.0023772930
Fugal_R_Inner	-	Fugal_R_Outer	3.7627516109	23	.0010120978

Note. Student's t-test.

Bayesian Paired Samples T-Test

Measure 1		Measure 2	BF_{01}	error %
Petal_Orthogonal	-	Fugal_Orthogonal	0.0000000107	$7.4172904801\times10^{-11}$
Petal_Radial	-	Fugal_Radial	0.0000399265	0.0000000012
Petal_O_Left	-	Petal_O_Right	4.3741932869	0.0242135392
Petal_O_Upper	-	Petal_O_Lower	0.6490304607	0.0000021662
Petal_O_Inner	-	Petal_O_Outer	0.0078014833	0.0002610459
Petal_R_Left	-	Petal_R_Right	3.5428517893	0.0251494331
Petal_R_Upper	-	Petal_R_Lower	0.3113874065	0.0000006691
Petal_R_Inner	-	Petal_R_Outer	0.0803404185	0.0000000707
Fugal_O_Left	-	Fugal_O_Right	4.1258734686	0.0245125528
Fugal_O_Upper	-	Fugal_O_Lower	0.0882494797	0.0000000740
Fugal_O_Inner	-	Fugal_O_Outer	0.0000973308	0.0000000007
Fugal_R_Left	-	Fugal_R_Right	4.0129009163	0.0246440069
Fugal_R_Upper	-	Fugal_R_Lower	0.0606153904	0.0000000777
Fugal_R_Inner	-	Fugal_R_Outer	0.0288163719	0.0000001202